

OPOYO

Pipeline note

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Abstract

OPOYO decides, from a live floor signal, whether a person-like impact occurred and whether the person then got up. This note is the claim set we have locked: what the pipeline does, where each number comes from, and how to collect the clips that will test the features. Labels on numbers are physics, Alwan range, runbook heuristic, placeholder, or unmeasured.

1 The job

The event we care about is a fall after which the person cannot rise and cannot call. Time on the floor is what produces the medical harm. That interval is called a long lie.

A wall button works only if the person can reach it and press it. A wearable works only if it is on and charged. A camera works in line of sight, and households refuse it in the bedroom and the bathroom. Those three paths leave this segment uncovered.

OPOYO reads the floor and the air, then waits to see whether anyone moves. The senior is not asked to press, wear, or speak.

2 What arrives at the laptop

The production node is a piezo and a microphone bonded to the slab. Classification runs on the device. The object that leaves the flat is a few bytes of event JSON: room, time, score, state.

The demo substitutes an iPhone for that node. The phone streams user-acceleration, already with gravity removed, in units of g (physics). The rate is 50 Hz (runbook heuristic). Each tick also carries one loudness number in dB. Packets are UDP JSON

$$\{v, id, model, t, ax, ay, az, mag, db\}$$

to port 9000 (runbook heuristic). The laptop multiplies magnitude by 9.81 to obtain m/s^2 (physics).

The phone is a sensor bridge. The slab measurement is a later substitution of the first stage. Downstream stages consume a clip and emit the same event object.

3 Why peak height is the wrong first question

A body and a dropped pan can put similar peak force into the floor. A detector that pages on peak height will page for a saucepan. The quantity that can separate those two events is shape: how long the envelope takes to fall, and where the energy sits in frequency.

That separation is a hypothesis about this floor. Histograms on labelled clips will confirm it or kill it. Until those histograms exist, the claim is unmeasured.

4 Pipeline

Processing is a sequence of maps. Each map has one job.

4.1 High-pass

A 2 Hz high-pass (runbook heuristic) removes gravity and slow drift from the magnitude trace. Walking, a television, and an impact still pass. The classifier later sees this high-passed trace, not the raw magnitude.

4.2 Trigger

The trigger asks whether the floor's energy just left the room's own quiet level. Let R be recent RMS values from frames treated as quiet. The threshold is

$$\theta = \text{median}(R) + k \cdot \text{MAD}(R).$$

The factor $k = 6$ is a runbook heuristic. Scaled MAD estimates σ when R is Gaussian (physics). A fire also requires the peak to exceed 3 m/s^2 (runbook heuristic). Quiet frames alone update the baseline, so a fall cannot raise the bar that should catch the next one. After a fire, a 2 s refractory period (runbook heuristic) blocks a second trip on the same ring-down. The trigger stays in warmup until five quiet samples have been seen (runbook heuristic).

This stage detects onset. It does not name the source.

A neighbour drilling is a second source in the same band. Median-plus-MAD assumes a roughly stationary quiet floor. Drilling violates that assumption. Crest factor would mark an impulse against a continuous buzz; the present trigger does not compute it.

4.3 Window

The decay that can separate a body from a click occurs after the peak. The peak is what the trigger just saw, so the tail does not yet exist. The extractor waits 1.5 s (runbook heuristic), then copies 0.5 s before the hit and 1.5 s after it. The resulting clip is 2 s long so the tail sits inside the window.

Our figure 6 caption draws a body decaying over about 1 s and a rigid object stopping within about 150 ms. Those times are unmeasured here. They are a schematic, not a citation.

Alwan et al. (2006) report a useful radius of about 15–20 ft on concrete (Alwan range). That bound is one construction. This build has not measured coverage on tile.

4.4 Seven features

Let w be the high-passed magnitude in the clip, and $e = |w|$. The rule classifier uses seven numbers from e .

Name	Definition	Job
peak	$\max e$	size of the hit
rms	$\sqrt{\text{mean}(w^2)}$	energy of the whole clip
crest	peak/rms	impulsiveness
rise_ms	time from $0.1 \cdot \text{peak}$ up to the peak	onset duration
decay_ms	time from the peak down to $0.1 \cdot \text{peak}$	tail duration
low_ratio	fraction of $ X(f) $ at low f	where the spectrum sits
centroid_hz	$\sum f X(f) / \sum X(f) $	the same, as one frequency

Crest, rise, and decay describe shape. Peak and RMS describe scale. Distance moves scale. Shape is what we are betting on.

The phone samples at 50 Hz (runbook heuristic), so Nyquist is 25 Hz (physics). Spectral features then see almost the whole band. Envelope features can still separate a bag from a book on hard tile. That separation is unmeasured until the histograms exist.

4.5 Score

Both classifiers implement $p = \text{score}(w) \in [0, 1]$. RuleClassifier is a sigmoid on decay, low-frequency share, and crest, and it runs with no weight file. ModelClassifier is a linear probe on the same seven numbers when `models/clf.joblib` exists.

The cut $p^* = 0.60$ is a placeholder. A precision–recall sweep on labelled files will replace it. Below the cut, the event is logged as `candidate` and the chain stops. Above the cut, the audio check may still suppress the alarm.

4.6 Audio check

A second classifier may delete an alarm the floor model proposed. If the check is off, broken, or given an empty clip, it returns “allow”. That is fail-open.

YAMNet wants about 16 kHz PCM (YAMNet input). The live phone sends one dB per tick. The demo check therefore uses a loudness envelope behind the same interface. The check does not require speech after the fall. Silence or a thud is an allowed clip. Television or sustained loudness is a blocked clip.

4.7 Confirm

A body-like, unblocked impact can still be a heavy sit. The machine then waits. The demo wait is 10 s (runbook heuristic). The product wait is 60 s (runbook heuristic). Motion in the window yields `recovered`. Cancel on the dashboard yields `cancelled`. Silence until the clock ends yields `alert`, once.

Only `alert` may send Telegram. The text carries room, local time, and score. A second message is scheduled 60 s later (runbook heuristic) unless someone acknowledges. `dry_run` defaults to true (runbook heuristic), so the send is a log line until credentials are filled.

5 Learning

There is no large labelled set of falls on this tile. A model trained on another floor will not transfer at the output layer, because damping changes decay. The extractor therefore stays frozen. Calibration fits only the last layer on this floor.

The intended set is about ten heel-drops and about ten dropped objects (runbook heuristic). Leave-one-file-out is the split: one physical event is one file. Report precision, recall, F1, and a confusion matrix. Accuracy is the wrong metric. Falls are rare, so a constant “no” looks good.

Whether the features work is empirical. If the decay histograms of heel-drop and book overlap on this floor, the theory does not hold here. Say that.

Misses and false alarms will remain. The product question is their rates per month. Those rates are unmeasured.

6 Distance

One sensor does not give range. Amplitude falls with distance (physics). A nearby tap and a far fall can share a peak. Coverage is measured at install, room by room. Alwan’s 15–20 ft on concrete (Alwan range) is a cited bound on one slab, not a specification for this unit.

Structure-borne waves in concrete travel on the order of 3–4 km/s (physics). Airborne sound travels at about 343 m/s (physics). Across five metres the air arrives about 13 ms later. At 50 Hz that offset is less than one sample. The demo therefore classifies two parallel clips with the phone’s own timestamps.

7 How to set the knobs

Tune from labelled files, after the histograms, before a live demo.

Knob	Now	Label	From data
k	6	runbook heuristic	raise if walk still fires; lower if heel-drop never fires
min_peak_mag	3 m/s ²	runbook heuristic	above walk, below heel-drop and bag
pre_s	0.5 s	runbook heuristic	long enough that rise is unclipped
post_s	1.5 s	runbook heuristic	long enough that decay is unclipped
p^*	0.60	placeholder	PR sweep on held-out files

If $k < 3$ is required to catch positives, the recordings are too weak. Move the phone closer and record again.

8 Work sequence

Build the pipeline. Collect labelled CSVs, one event per file. Plot `decay_ms`, `crest`, and `low_ratio` on this floor. Fit the last layer if the classes separate. Test on unseen takes: heel-drop and bag should reach `alert` or `recovered`; book, walk, and quiet should not send Telegram.

Direction of the hit is unused in this binary proof of concept.

A Collection

A friend can collect without the dashboard. Stop unicorn first: recorder and dashboard both bind UDP 9000.

Hard floor, phone face-down, case off, hotspot. Keep distance fixed at 1.5 to 2 m. From the `opoyo-ellipsis` directory:

```
python scripts/record.py --interactive --floor tile --room kitchen
```

Each take: two seconds still, one action, two seconds still, stop. A new file is a new take.

Positives: ten heel-drops (toes, then both heels, once) and ten bag-drops (backpack 8–12 kg, flat, from about waist height). A jump leaves the floor, so it is a different signal.

Negatives: book $\times 5$, pan $\times 5$, door slam $\times 8$, walk five paces $\times 8$, quiet 15 s $\times 2$. Optional: television on, no impact, 10 s $\times 2$.

The recorder writes `data/<label>_<nn>.csv` with columns *t*, *ax*, *ay*, *az*, *mag*, *db*, and appends one row to `data/metadata.csv`. Required metadata: floor, room, distance, case, object. Send the `data/` folder.

Without a phone, in another window:

```
python scripts/fake_phone.py --impact heeldrop --seconds 8
```